

MSE 311 Thermodynamics of materials-Homework 14

Name:

Student ID:

1. When a weight was hung on a pure Cu wire of radius $r = 0.1$ mm and held at a high temperature, the balanced load of the weight was $m = 5.5 \times 10^{-5}$ kg, which caused neither expansion nor shrinkage of the wire. Estimate the surface energy of pure Cu from this experimental value.

2. Steel can be hardened and made more corrosion resistant by heat treating in a nitrogen atmosphere to diffuse nitrogen into the surface layer of the steel. This process can be roughly modeled using error function solution as following.

$$C(y, t) = A + B \operatorname{erf}\left(\frac{y}{\sqrt{4Dt}}\right)$$

Assume that a plate of steel is heat treated in a nitrogen environment at 500°C , where the solubility of nitrogen in steel is $C_N^* \approx 0.5$ mol % and the initial background concentration of nitrogen in the steel is zero ($C_N^0 = 0$).

- Provide the boundary condition and initial condition for this problem.
- Based on your boundary and initial conditions, provide the A and B of error function solution in this problem.
- Assuming the diffusivity of nitrogen in steel is $D_{N/\text{steel}} = 10^{-9}$ cm²/s at the heat treatment temperature, determine how long it will take for the nitrogen concentration 10 μm deep into the steel to reach 70% of its value on the surface (i.e., 70% of C_N^*)?
- How long would it take the nitrogen concentration 20 μm deep into the steel to reach 70% of its value on the surface?

3. Estimate the following parameters in solidification of pure water by the values in following Table

1. (a) the change of Gibbs energy of melting (ΔG_V), (b) critical nucleation size (r^*) and the change of Gibbs energy to form a ordering crystalline cluster in a diameter of r^* (ΔG_r^*), (c) frequency of atomic transition (ν_c) and homogeneous nucleation rate (I). The diffusion coefficient of the liquid phase is presumed as $D^L \approx 10^{-9}$ m²/s. In the case of ice, the volume of a molecule is 3×10^{-29} m³.

Table 1 Calculated values for parameters of water regarding solidification

Substance	Melting point (T_m), K	Latent heat of fusion (L_V), kJ/mol	Molar volume (V), m ³ /mol	Interface energy (γ), J/m ²	Maximum degree of supercooling (ΔT_{max}), K
H ₂ O	273	6.0	18×10^{-6}	0.025	40

4. The nucleation process of pure metal solidification usually requires a large degree of supercooling, but the concept of "superheat" is rarely mentioned when pure metal is melted. Why?

5. Figure 1 gives a set of rate of transformation curves for the crystallization of a glass. Using this information, calculate and plot a TTT (time–temperature–transformation) diagram for this crystallization that includes curves for 10, 50, and 90% transformed with temperature on the vertical axis versus log (time) on the horizontal axis.

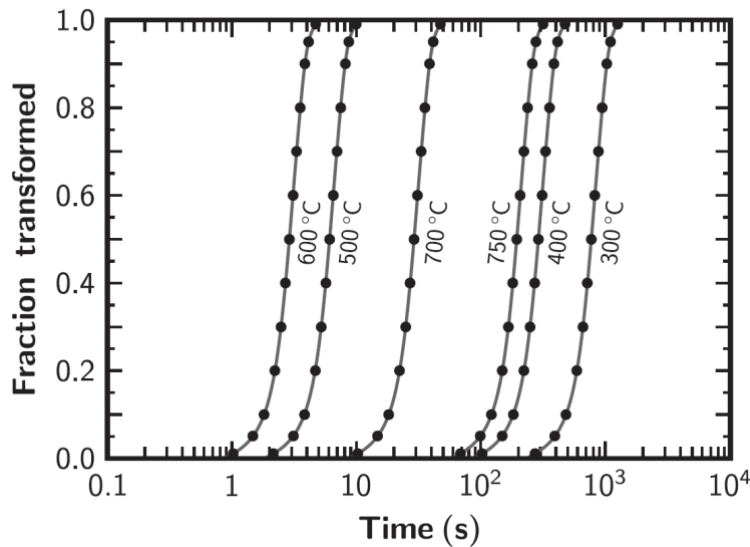


Figure 1 Crystallization of a glass as a function of time for various transformation temperatures.

6. Al_2O_3 , which melts at 2324 K, and Cr_2O_3 , which melts at 2538 K form complete ranges of solid and liquid solutions. Assuming that $\Delta S_{m, \text{Cr}_2\text{O}_3}^\circ = \Delta S_{m, \text{Al}_2\text{O}_3}^\circ$, and that the solid and liquid solutions in the system $\text{Al}_2\text{O}_3\text{-Cr}_2\text{O}_3$ behave ideally. $\Delta H_{m(\text{Al}_2\text{O}_3)}^\circ = 107.5 \text{ kJ/mol}$. When an alloy of $X_{\text{Al}_2\text{O}_3} = 0.5$ is heated, the temperature at which equilibrium melting is beginning and complete is 2420K and 2444K, respectively. Calculate

(1) $\Delta G_{m(\text{Al}_2\text{O}_3)}^\circ$ and $\Delta G_{m(\text{Cr}_2\text{O}_3)}^\circ$ in terms of temperature (T).

(2) The composition of the melt which first forms.